Claire Kim

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PhD Candidate in Computer Science with industry experience building LLM-powered systems, evaluation frameworks, and scalable ML pipelines. Developed production-facing AI solutions at Amazon and AMD, including RAG systems, synthetic data generation, LLM-as-a-judge evaluation, and multi-agent workflows. Research focuses on trustworthy AI, reasoning, and human-AI decision support.

EDUCATION

ARIZONA STATE UNIVERSITY — **2021 – Aug 2026**
PhD, Computer Science — Tempe, AZ
- Data Mining & Machine Learning Lab
- Advisors: Dr. Huan Liu and Dr. Mickey Mancenido
- Funded by DHS-CAOE

KOREA UNIVERSITY — **2013 – 2019**
M.E., Computer Science & Engineering — Seoul, South Korea
B.S., Computer Science & Engineering

WORK EXPERIENCE

AMAZON — **Sep – Dec 2025**
Applied Scientist Intern — **Bellevue, WA**
- Improved production sentiment forecasting models supporting community operations at Amazon by developing advanced sequence modeling approaches and improving prediction robustness at scale.
- Designed and deployed an end-to-end ML pipeline spanning data collection, model training, experiment tracking, and production deployment using Python, AWS, MLflow, and CI/CD workflows for reliable model releases.

AMD — **May – Aug 2025**
AI/ML Intern — **Austin, TX**
- Developed Q-RAG, a question-centric retrieval framework that uses synthetic QA generation and LLM-as-a-judge evaluation to identify knowledge gaps and reduce hallucinations in enterprise AI systems, achieving up to 87% agreement with domain experts in automated QA validation.
- Built scalable synthetic QA generation pipelines and automated evaluation workflows, enabling benchmarking of RAG systems without extensive expert annotation.

AMD — **Aug – Dec 2024**
Software Development Intern — **Austin, TX**
- Designed LLM inference and evaluation infrastructure supporting RAG-infused multi-agent workflows, user-feedback integration, and large-scale experimentation.
- Optimized data and inference pipelines for efficiency, improving throughput and reducing evaluation latency.

DHS-CAOE (Center for Accelerating Operational Efficiency) — **May 2022 – Aug 2024**
Graduate Research Assistant — **Tempe, AZ**
- Designed NLP models for topic modeling and text summarization using BERT and Llama-2.
- Collaborated with an interdisciplinary team to build a multi-agent RAG system for trustworthy AI-enabled intelligence analysis, validated via human-subject study using GPT-4.
- Developed an interactive data analysis and visualization dashboard using NodeJS and Flask.

ONR (Office of Naval Research Project) — **Jan 2021 – Aug 2022**
Graduate Research Assistant, Arizona State University — **Tempe, AZ**

- Researched the integration and mutual influence of online and offline COVID-19 datasets using topic modeling.
- Analyzed 2M tweets for sentiment and stance detection in pandemic-related discussions.

SELECTED PROJECTS

Reasoning-Level Fairness in LLMs (2026)

- Developing evaluation methods for fairness in multi-step reasoning and chain-of-thought generation.
- Investigating bias propagation across inference trajectories and agent workflows. Based on dissertation work exploring reasoning-level fairness in human-AI systems.

TECHNICAL SKILLS

- **Large Language Models & LLMOps:** AI Safety, Model Alighment, Post-training, Retrieval Augmented Generation, Agentic AI, Multi-Agent Systems, Evaluation Frameworks, LLM-as-a-Judge, Synthetic Data Generation, Prompt Engineering, RLHF, RLAIF, LoRA, PEFT, Inference-Time Scaling, Responsible AI
- **Machine Learning:** PyTorch, TensorFlow, Scikit-Learn, Hugging Face, Pandas, NumPy
- **Infrastructure & MLOps:** AWS, GCP, Docker, MLflow, WandB, CI/CD, TensorRT, Model Serving, GPU Optimization
- **Retrieval & Search:** FAISS, Weaviate, Chroma, Vector Search, Knowledge Retrieval
- **Software Engineering:** Flask, Node.js, Git, Linux, REST APIs
- **Languages:** Python, SQL, Java, JavaScript, Bash
- **AI-assisted dev:** Claude Code, Codex, GitHub Copilot, agentic coding workflows, MCP

SELECTED PUBLICATIONS | Google Scholar

- **AI Magazine'25** PADTHAI-MM: A Principled Approach for the Design of Trustworthy, Human-Centered AI systems using the MAST Methodology
- **ASONAM'24** Robust Stance Detection: Understanding Public Perceptions in Social Media
- **JAIR'23** Evaluating Trustworthiness of AI-Enabled Decision Support Systems: Validation of the Multisource AI Scorecard Table (MAST)
- **COLING'22** Debiasing Word Embeddings with Nonlinear Geometry
- **SBP-BRiMS'22** Bridge the Gap: the Commonality and Differences Between Online and Offline COVID-19 Data

ACADEMIC SERVICE & TEACHING

- Reviewer at AMLC 2025 (Amazon Gen AI Evaluation Workshop) 2025
- Invited Reviewer for NeurIPS 2025 (Reliable ML Workshop) 2025
- Program Committee (PC) member of ASONAM 2024 conference 2024
- Invited talk at Machine Learning Day 2024, Arizona State University 2024
- Invited talk at SCAI AI Day, Arizona State University 2023
- Program Committee (PC) member of ASONAM, SBP-BRiMS 2023 conferences 2023
- Invited Reviewer for EMNLP 2023 conference 2023
- Reviewer at ECML-PKDD, ACM MultiMedia, ASONAM, and AAAI conferences 2022
- Graduate Teaching Assistant, Object-Oriented Programming & Data Structures (ASU) 2022